Impact of International Trade, Minimum Wage, and GDP Growth on Youth Employment

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Abstract

This paper looks at how youth employment in the United States has been affected by GDP growth, international trade, and real minimum wage from 1991 to 2024. Using a regression model, the study focuses on people aged 15 to 24. The results show that when the economy grows, youth employment also increases. On the other hand, a rise in trade is linked to a small drop in youth employment. Changes in the real minimum wage do not seem to have a clear effect. These findings suggest that overall economic growth and trade are more important than wage policies in shaping job opportunities for young people.

I. Introduction

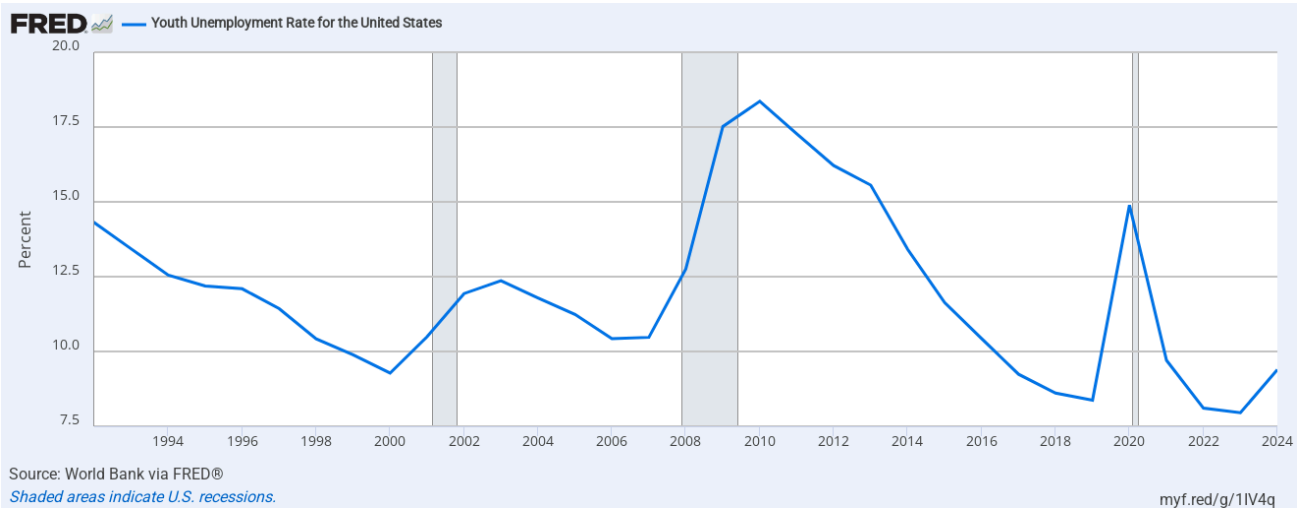
The cost of living in the United States has gone up a lot over the past few years. Prices for most consumer items have jumped more than 20% (Marfice, 2024), and that has put a real financial strain on many families. Young people, especially, have felt the pressure, with many stepping into the workforce earlier than they planned to help support their households.

Even though more young people are looking for work, finding a steady job is still difficult. Youth unemployment continues to be a serious issue. Young workers often don't have enough experience, the right professional connections, or degrees that match what employers are looking for. That's why it's important to understand what factors are helping or hurting youth employment today.

This paper looks at three important economic factors that could be shaping youth employment: GDP growth, international trade, and minimum wage policies. These forces have real effects on job opportunities. When American companies export more, production usually goes up, leading to more jobs. But when imports of cheaper goods rise, local industries sometimes suffer, which can result in fewer jobs, especially for young workers starting out. Changes in minimum wage laws can also play a role; higher wages can improve living standards, but might also discourage businesses from hiring inexperienced workers.

The goal of this study is to figure out how GDP growth, imports, exports, and minimum wage changes impact youth employment across the U.S. Using data from 1991 to 2024, the results show that GDP growth has a strong positive impact on youth employment. On the other hand, higher trade volumes are linked to a decrease in youth employment. Changes in the real minimum wage did not seem to have a major effect either way. These findings help explain some of the challenges and opportunities young workers face in today's economy.

GRAPH 1- Youth Unemployment Rate for the United States, Dating from 1991 to 2024



As shown in Graph 1, the youth unemployment rate has seen noticeable spikes, particularly during the 2008 financial crisis and again during the COVID-19 pandemic in 2020, when the rate sharply increased. These periods highlight how sensitive young workers are to broader economic downturns.

Even in times of recovery, youth unemployment doesn't always bounce back quickly, making it important to understand what underlying factors are at play.

II. Literature Review

Existing research shows that international trade and minimum wage policies can each shape youth employment, although their combined effects are less explored. Many studies have looked at how these factors interact with larger economic trends to either create opportunities or present obstacles for young workers.

(Kalenkoski, 2024) studies how minimum wage policies affect youth employment, unemployment, and income. The research shows that raising the minimum wage can increase earnings for employed youth, but it can also reduce job opportunities for others, particularly unskilled young workers. Higher minimum wages can lead to delayed entry into the labor market and fewer chances for on-the-job training, which could lower lifetime earnings. The study highlights the importance of designing minimum wage policies carefully to balance the benefits of higher pay against the risk of job loss.

(Olabisi, 2020) looks at the effects of international trade on youth jobs in his study "Trade Shocks and Youth Jobs." He finds that young workers are hit harder by trade shocks than older workers. Sectors facing more import competition show significantly higher job losses among workers under 25. In fact, for every job lost by an older worker, about three jobs are lost by young workers. Although exports can create new jobs, the study points out that the overall supply chain effects matter because job losses spread to related industries, too.

A recent report from the International Labour Organization shows that increases in imports usually hurt job growth and wages for young workers (International Labour Organization, 2024). It shows that increases in imports usually hurt job growth and wages for young workers, while export growth can create new opportunities. However, many of these new jobs demand skills that younger workers often lack, which emphasizes the need for better education and vocational training.

Altogether, the literature paints a complicated picture: minimum wage hikes can have mixed effects depending on how they are set, and trade flows can either open or close doors depending on industry and worker skill levels. These findings highlight the need for balanced economic policies that protect young workers while encouraging economic growth. Building on this foundation, this paper explores how trends in trade, wage policies, and GDP growth have been shaping youth employment outcomes in the United States in recent decades.

III. Description of the Model

The model analysed in this paper is:

$$youthemp_i = \beta_0 + \beta_1 GDPgrowth_i + \beta_2\, ln(import)_i + \beta_3\, ln(export)_i + \beta_4\, realminwage_i + \varepsilon_i$$

Youth employment (*youthemp*) serves as the dependent variable (Y), and GDP growth (*GDPgrowth*) is the primary independent variable (X_1). Imports (*import*), exports (*export*), and real minimum wage(*realminwage*) are additional independent variables that account for external economic factors affecting youth employment outcomes.

GDP growth is the focus because when the economy grows, businesses tend to expand and hire more workers, which can create more entry-level jobs for young people. Higher GDP growth usually means more opportunities across different industries. Imports are included because when a country brings in more goods from abroad, it can hurt local industries that employ young workers. Exports, on the other hand, are important because selling more goods to other countries can boost production and create jobs at home. Minimum wage is added because changes in wage laws can affect whether businesses decide to hire more young workers, especially for part-time or seasonal jobs where youth employment is high.

By including all these factors in the model, the goal is to better understand what drives youth employment without leaving out important influences. Using these variables helps avoid bias and gives a clearer picture of each factor's impact. The data used covers all 50 U.S. states from 1991 to 2024, providing a long-term view that captures economic ups and downs.

The main research question is whether and how GDP growth, international trade (imports and exports), and minimum wage levels affect youth employment outcomes in the United States. It is expected that GDP growth will be positively related to youth employment, exports will support job creation, imports may be associated with job losses, and minimum wage effects may vary depending on economic conditions and the size of the changes.

The hypothesis for GDP growth is as follows: the null hypothesis (H_0) states that GDP growth does not affect youth employment ($\beta_1 = 0$). The alternative hypothesis (Ha) states that GDP growth does affect youth employment ($\beta_1 \neq 0$).

IV. Data and Analysis

A. Data Sources and Summary

Table 1. Codebook for Youth Employment, GDP Growth, Import, Export, and Minimum Wage.

Variable Name	Description	Type	Year	Source
youthemp	The youth employment rate measures the proportion of young people who are employed.	Dependent	1991-2024	WorldBank
GDPgrowth	Percentage increase in gross domestic product (GDP) over a period.	Independent(main)	1991-2024	FRED
ln(import)	The value of goods and	Independent	1991-2024	FRED

	services brought into the country.			
ln(export)	The value of goods and services sold to other countries	Independent	1991-2024	FRED
realminwage	Minimum wage adjusted for inflation, representing the actual purchasing power of the legally mandated lowest hourly pay. It is calculated by dividing the nominal minimum wage by a price index (such as the Consumer Price Index), allowing comparisons of minimum wage levels over time in terms of what workers can buy.	Independent	1991-2024	FRED

Youth employment rate data from 1991 to 2024 was collected from the World Bank, tracking the percentage of young people aged 15-24 employed across all 50 U.S. states. GDP growth rates by state were pulled from the FRED database, showing the yearly percentage change in economic output. Import and export values were also taken from FRED and adjusted using natural logarithms to make the data easier to work with and reduce skewness. For real minimum wage data, I gathered the state-level nominal minimum wage for each year, averaged the values across all states, and then adjusted the figures for inflation using the Consumer Price Index (CPI). Altogether, this dataset, covering more than three decades, provides a strong base to analyze how GDP growth, trade, and minimum wage changes have shaped youth employment outcomes in the United States.

Table 2. Descriptive Statistics

Variable	**Observations (N)**	**Mean**	**Median**	**Minimum**	**Maximum**	**Standard Deviation**
youthemp	34	47.99	47.73	40.97	54.35	4.210192
GDPgrowth	34	2.518	2.723	-2.576	6.055	1.755035
ln(import)	34	7.547	7.750	6.435	8.315	0.550128
ln(export)	34	7.309	7.391	6.388	8.065	0.508672

realminwage	34	9.780	9.435	8.420	11.400	0.9399371

34 observations correspond to years from 1991 to 2024. The average youth employment rate is 47.99%, ranging from 40.97% to 54.35%. GDP growth averages 2.52% but includes both negative and positive values, reflecting periods of contraction and expansion. The log-transformed values for imports and exports show moderate variation, with means of 7.55 and 7.31, respectively. The real minimum wage averages $9.78, with values between $8.42 and $11.40.

B. Analysis of Multivariate Regression Model

Table 3: Multivariate Regression - Estimated Model

	Coefficient estimate	**Standard error**	**t value**	**p value**
Constant	91.1278	6.5434	13.927	0.0000000000000224***
GDPgrowth	0.7823	0.2328	3.361	0.00219 **
ln(import)	2.0910	4.2915	0.487	0.62975
ln(export)	-8.0038	5.5286	-1.448	0.15842
realminwage	-0.2444	1.0392	-0.235	0.81572
	(N)		(R^2)	
	34		0.7352	

*p < 0.05, **p < 0.01, ***p <0.001

Looking at the results, GDP growth has a clear link to youth employment. The coefficient is 0.7823, which tells us that if GDP rises by 1 percentage point, youth employment goes up by roughly 0.78 percentage points, assuming everything else stays the same. The t-statistic here is 3.361, which is higher than the critical value of 2.045 at the 5% level with 29 degrees of freedom. Since the p-value is 0.00219, this result is statistically significant. In simple terms, stronger economic growth tends to create more jobs for young people. For example, after the Great Recession, Texas bounced back faster than many other states. Its steady growth helped boost hiring in areas like retail and services, where young workers often find their first jobs (Texas State Historical Association, n.d.).

The model incorporates trade variables, such as the logarithm of imports and exports, to evaluate the impact of trade activity on youth employment. Given that these variables are expressed

in logarithmic form, their coefficients indicate the shift in youth employment (in percentage points) resulting from a one percent change in trade volume.

The coefficient for imports is 2.0910. This indicates that a one percent rise in import volume correlates with a 2.09 percentage point rise in youth employment. Nonetheless, this outcome does not hold statistical significance. The t-statistic stands at 0.487, which is significantly lower than the critical value, with a p-value of 0.62975. This suggests we cannot assert with confidence that alterations in imports influence youth employment. A potential explanation for the positive coefficient might be that certain imported consumer products boost activity in retail, logistics, or warehousing, potentially helping younger workers indirectly. However, due to this elevated p-value, we are unable to make trustworthy conclusions.

The coefficient for exports is -8.0038, which means that a one percent increase in exports is linked to about a 0.08 percentage point drop in youth employment. However, this finding isn't statistically significant, since the t-statistic is -1.448 and the p-value is 0.15842, both of which fall outside acceptable thresholds. The negative sign might seem unexpected, but it could be explained by the nature of industries that focus on exports. Sectors like tech, aerospace, or advanced manufacturing often require specialized training and tend to hire experienced workers instead of younger or less skilled ones. As a result, even if exports grow, the benefits may not reach the youth labor market directly.

It is important to note that the high correlation between imports and exports may be inflating the standard errors of both estimates. This condition, known as multicollinearity, makes it difficult to separate the individual impact of each trade component. Because of this, a revised model is tested later using a combined trade volume variable to address the issue.

The real minimum wage coefficient is -0.2444, suggesting that a one-dollar increase in the inflation-adjusted minimum wage is associated with a 0.24 percentage point decline in youth employment. However, the t-statistic is -0.235, and the p-value is 0.81572, which clearly indicates no statistical significance. This aligns with mixed findings in the literature. While higher wages may slightly reduce hiring incentives for entry-level roles, other factors like local demand, training programs, or employer size may have a stronger effect.

Overall, the model fits the data fairly well. The R-squared value is 0.7352, which means it accounts for about 73.5 percent of the changes in youth employment over time. Still, GDP growth is the only variable that turns out to be statistically significant. While trade and minimum wage are relevant topics, their effects aren't strong enough in this model to draw firm conclusions. To deal with some of the issues, like overlapping effects between imports and exports, the next part of the paper takes a different approach by combining them into one trade measure.

Table 4: Multivariate Regression With New Variable

	Coefficient estimate	**Standard error**	**t value**	**p value**
Constant	91.3451	6.7571	13.518	0.0000000000000269 ***
GDPgrowth	0.7691	0.2332	3.298	0.00251 **
lmtradevolume	-4.5087	1.5296	-2.948	0.00615 **
realminwage	-0.8835	0.8644	-1.022	0.31492
	(N)		(R^2)	
	34		0.7243	

*p < 0.05, **p < 0.01, ***p <0.001

To deal with the issue of multicollinearity between imports and exports, I created a new variable by combining the natural logs of both. This combined trade variable, called lmtradevolume, gives a general picture of trade activity while reducing the overlap between the original variables. The revised model now includes GDP growth, real minimum wage, and this new trade variable.

In this updated model, GDP growth still has a clear and positive relationship with youth employment. The coefficient is 0.7691, which means that a one percentage point increase in GDP growth is associated with roughly a 0.77 percentage point rise in youth employment. With a t-statistic of 3.298 and a p-value of 0.00251, the result is statistically significant at the 1% level. This supports the idea that when the economy grows, more jobs open up for young workers.

The coefficient for the combined trade variable is -4.5087. This suggests that a one percent increase in overall trade volume is linked to about a 0.045 percentage point drop in youth employment. The t-statistic here is -2.948, and the p-value is 0.00615, so the result is statistically significant. Compared to the earlier model, where the separate trade variables gave unclear results, this version tells a more consistent story. One possible reason for the negative sign is that the jobs created by trade are not always accessible to younger workers. Sectors like manufacturing and tech, which are heavily trade-dependent, often seek more experienced labor or move lower-skilled jobs overseas. As a result, young people may not benefit as much from increased trade activity.

As for the real minimum wage, it remains statistically insignificant. Its coefficient is -0.8835, meaning a one-dollar increase is associated with a small decline in youth employment. But the t-statistic is just -1.022, and the p-value is 0.31492, so the result is not strong enough to draw any clear conclusions. It's likely that differences across states, such as how the minimum wage is enforced or what local job markets look like, contribute to the unclear impact in this national dataset.

The R-squared value for this revised model is 0.7243, which means that the variables together explain about 72.4% of the variation in youth employment. While this is slightly lower than

the earlier model's 73.5%, the improvement comes from the clearer role of trade. GDP growth and trade volume both show up as significant predictors of youth employment, while the minimum wage still does not. By using a combined trade measure, this model gives a better sense of how overall trade patterns relate to the job market for young people.

C. F-Test For Import, Export, and GDP Growth

Since GDP growth is closely related to international trade activity, particularly through changes in imports and exports, it is important to test whether these three variables, GDP growth, log(imports), and log(exports), are jointly significant in explaining youth employment. To do this, I conducted an F-test to evaluate the following hypotheses:

Null hypothesis (H_0): β_GDPgrowth = β_log(import) = β_log(export) = 0
Alternative hypothesis (Ha): At least one of these coefficients is not equal to zero

Applying the F-test formula:

$$F = \frac{(R^2_{unrestricted} - R^2_{restricted})/q}{(1 - R^2_{unrestricted})/(n-k-1)}$$

Where $q = 3$ (the number of restrictions), $n = 34$ (sample size), and $k = 4$ (number of regressors in the unrestricted model), the calculation proceeds as follows:

$$F = \frac{(0.7352 - 0.5482)/3}{(1 - 0.7352)/(34-4-1)} = \frac{0.187/3}{0.2648/29} = \frac{0.0623}{0.00913} \approx 6.82$$

The critical value for the F-distribution at the 5% significance level with 3 and 29 degrees of freedom is approximately 2.95. Since the calculated F-statistic of 6.82 exceeds this threshold, we reject the null hypothesis that the coefficients on GDP growth, log(imports), and log(exports) are jointly zero.

This result demonstrates that GDP growth and trade activity-measured by imports and exports-are jointly significant in explaining youth employment outcomes. The implication is that macroeconomic and trade conditions together play a substantial role in shaping job opportunities for young people. For example, while GDP growth can generate new jobs, an increase in imports may offset these gains by adversely affecting local industries that tend to employ younger workers. Conversely, export growth can support employment, but the new jobs created may often require skills that younger workers lack. These findings underscore the complex interplay between economic growth and trade in determining youth employment, highlighting the importance of considering these factors together rather than in isolation.

V. Limitation

This study has a few limitations worth noting. First, it uses national-level data, which may hide important differences across states. For example, youth employment trends in a manufacturing-based state like Michigan might look very different from those in states with more service-based economies. Second, the trade variable combines imports and exports to deal with multicollinearity, but this may blur the different ways imports and exports affect jobs.

Third, other factors could influence youth employment that weren't included in the model. Things like education levels, technology access, or the types of industries present in different areas could all play a role. Lastly, while minimum wage didn't appear to have a significant effect nationally, that doesn't mean it has no impact. State-level differences in wage policies, enforcement, or cost of living might lead to different outcomes. Future studies could explore these local differences in more detail.

VI. Conclusion

This study emphasizes the significance of macroeconomic factors in influencing youth employment results in the United States. The results show that GDP growth positively and significantly impacts youth employment, highlighting the importance of widespread economic development. Conversely, an increase in total trade volume decreases youth employment, potentially because of changing labor needs or competitive strain in low-skill industries. National changes in minimum wage did not demonstrate a significant effect.

Nonetheless, additional research into regional impacts is necessary, considering the differences in wage policies at the state level. This research highlights the importance of implementing growth-focused policies and also promotes more in-depth, localized studies on the unique issues confronting young workers in various regions of the nation. Future policy initiatives ought to take into account access to education, specialized training for various industries, and regional economic factors to better aid youth employment across the country.

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